ICT702 Data Visualisation

Data Visualisation and Design

Module 3

USC

I would like to acknowledge the traditional custodians of the land we are meeting on,

The Gubbi Gubbi/Kabi Kabi peoples,

and pay my respect to Elders past, present and emerging.

Part 2

Gestalt Principles

Gestalt principles refer to the guiding principles of how people interpret and perceive what they see.

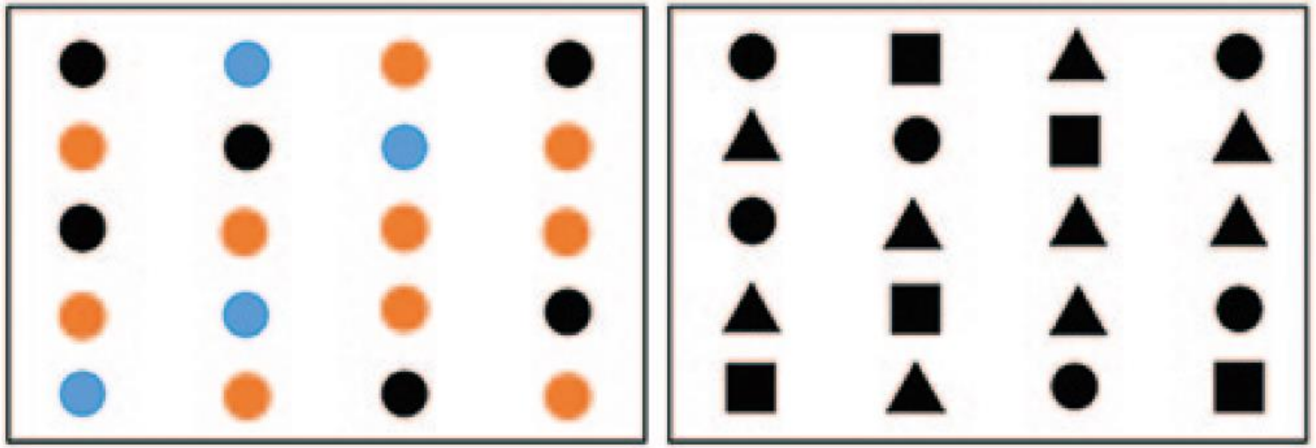
- These principles generally describe how people define order and meaning in the things they see.

The four main Gestalt principles related to the design of data visualizations are:

- **Similarity**
- **Proximity**
- **Enclosure**
- **Connection**

Gestalt Principle of Similarity

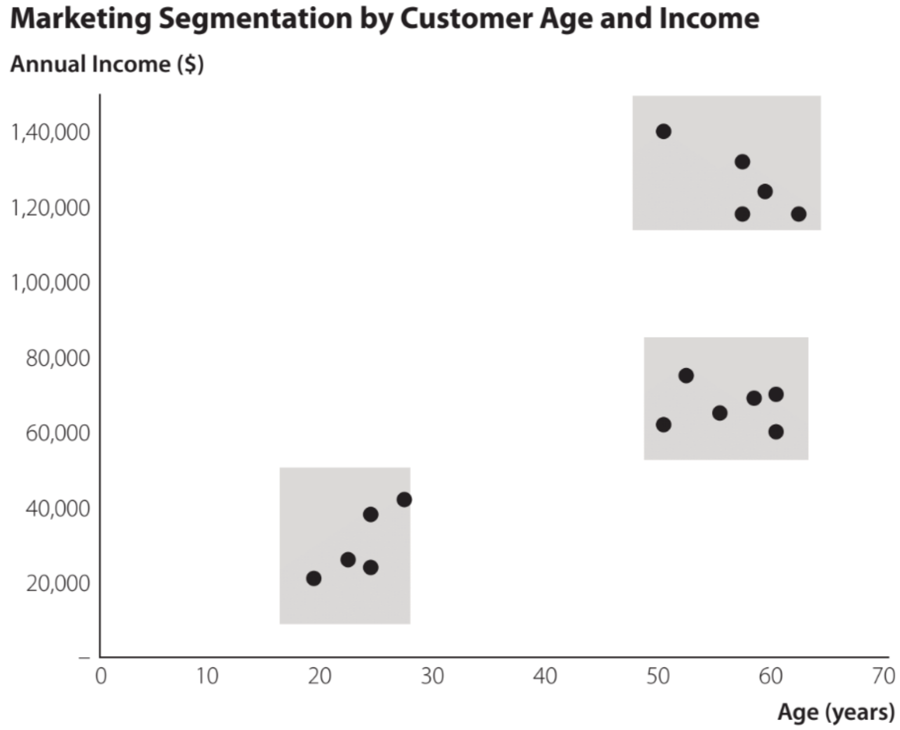
People consider objects with similar characteristics as belonging to the same group



The audience will perceive objects of the same colour, or the same shape, as belonging to the same group.

Gestalt Principle of Proximity

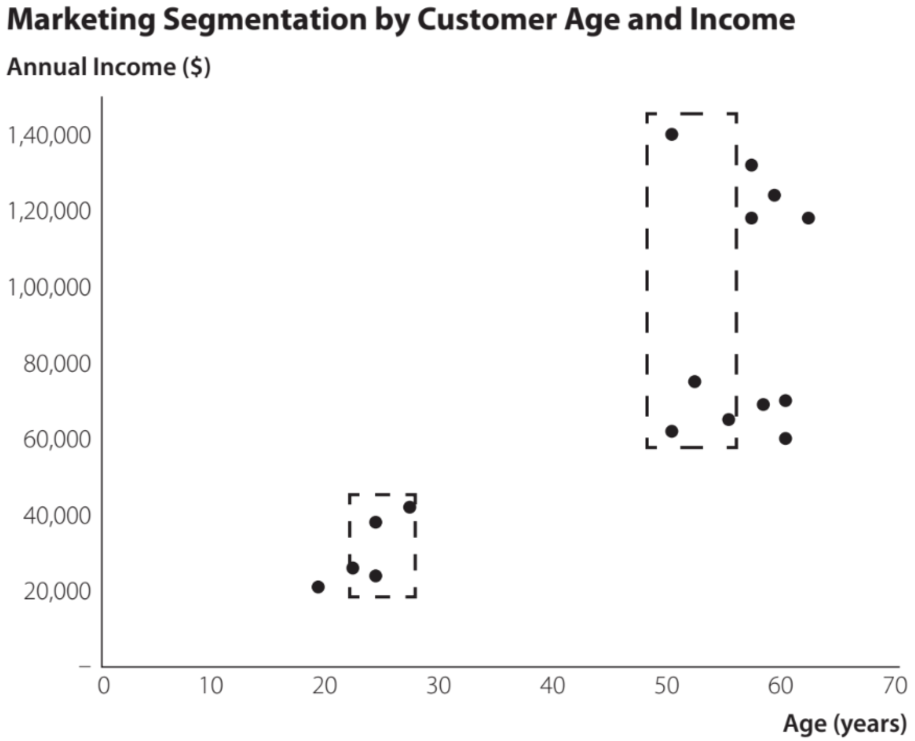
People consider objects physically close to one another as belonging to the same group



People will seek to group objects near each into the same group and objects that are far from one another into different groups.

Gestalt Principle of Enclosure

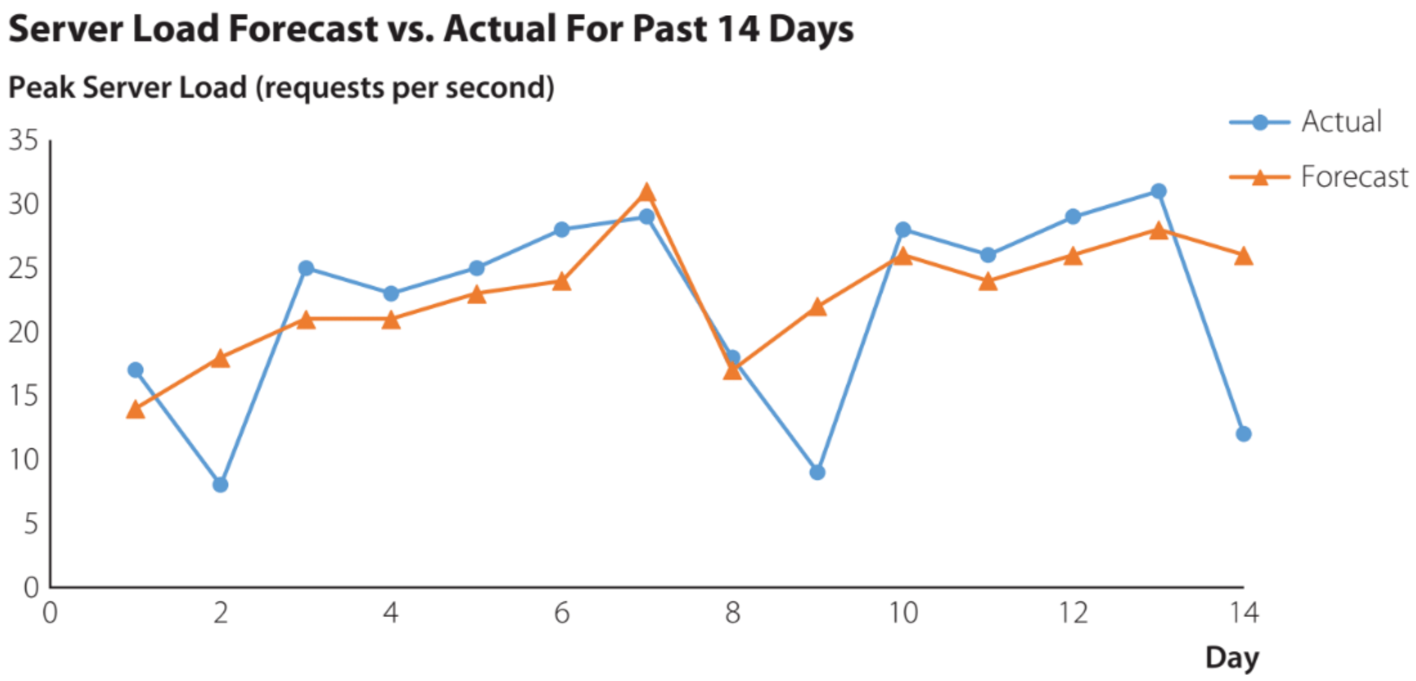
People consider objects physically enclosed together as belonging to the same group



The presence of a third attribute besides age and income, such as educational background, can be visually indicated with enclosures.

Gestalt Principle of Connection

People interpret connected objects as belonging to the same group



The principle of connection in data visualization is often used for time-series data.

Connecting the markers emphasizes trends and makes it easier to separate forecast values from actual values.

Data-Ink Ratio

- **Data-ink ratio** is the proportion of ink used for data to the total amount of ink in a table or chart.
  - Helpful for creating effective tables and charts for data visualization
  - **Data-ink**: ink used in a table or chart that is necessary to convey the meaning of the data to the audience
  - **Non-data-ink**: ink used in a table or chart that serves no useful purpose in conveying the data to the audience
  - **Decluttering** is the process of increasing the data-ink ratio in a chart

Data-Ink Ratio in Tables (1 of 2)

Example of low data-ink table

Scarf Sales			
Day	Sales (units)	Day	Sales (units)
1	150	11	170
2	170	12	160
3	140	13	290
4	150	14	200
5	180	15	210
6	180	16	110
7	210	17	90
8	230	18	140
9	140	19	150
10	200	20	230

Example of high data-ink table

Scarf Sales			
Day	Sales (units)	Day	Sales (units)
1	150	11	170
2	170	12	160
3	140	13	290
4	150	14	200
5	180	15	210
6	180	16	110
7	210	17	90
8	230	18	140
9	140	19	150
10	200	20	230

White space is the portion of a data visualization that is devoid of markings. Its use is equivalent to increasing the data-ink in a visualization.

Different Levels of Data-Ink Ratio in Tables

Comparing different table designs with different data-ink ratios

Design A:

	Month						
	1	2	3	4	5	6	Total
Costs ($)	48,123	56,458	64,125	52,158	54,718	50,985	326,567
Revenues ($)	64,124	66,128	67,125	48,178	51,785	55,687	353,027
Profits ($)	16,001	9,670	3,000	(3,980)	(2,933)	4,702	26,460

Design B:

	Month						
	1	2	3	4	5	6	Total
Costs ($)	48,123	56,458	64,125	52,158	54,718	50,985	326,567
Revenues ($)	64,124	66,128	67,125	48,178	51,785	55,687	353,027
Profits ($)	16,001	9,670	3,000	(3,980)	(2,933)	4,702	26,460

Design C:

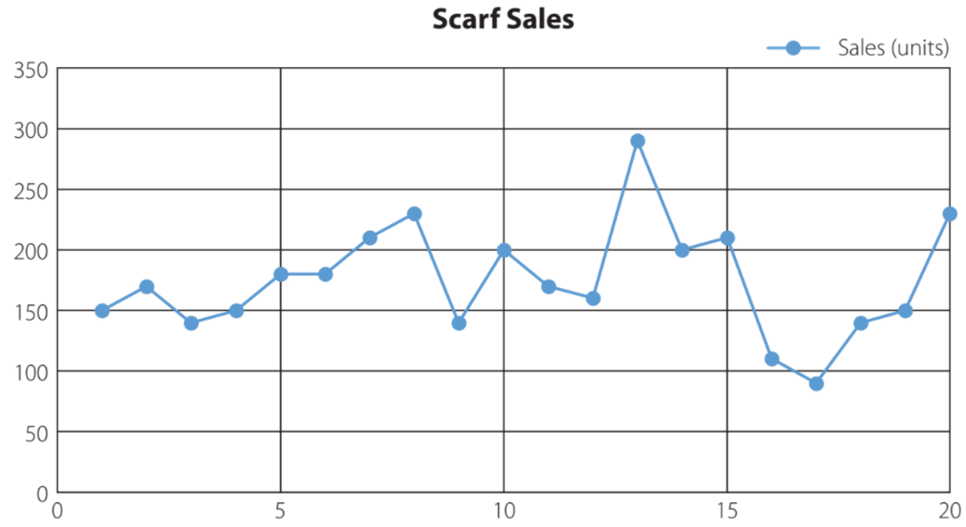
	Month						
	1	2	3	4	5	6	Total
Costs ($)	48,123	56,458	64,125	52,158	54,718	50,985	326,567
Revenues ($)	64,124	66,128	67,125	48,178	51,785	55,687	353,027
Profits ($)	16,001	9,670	3,000	(3,980)	(2,933)	4,702	26,460

Design D:

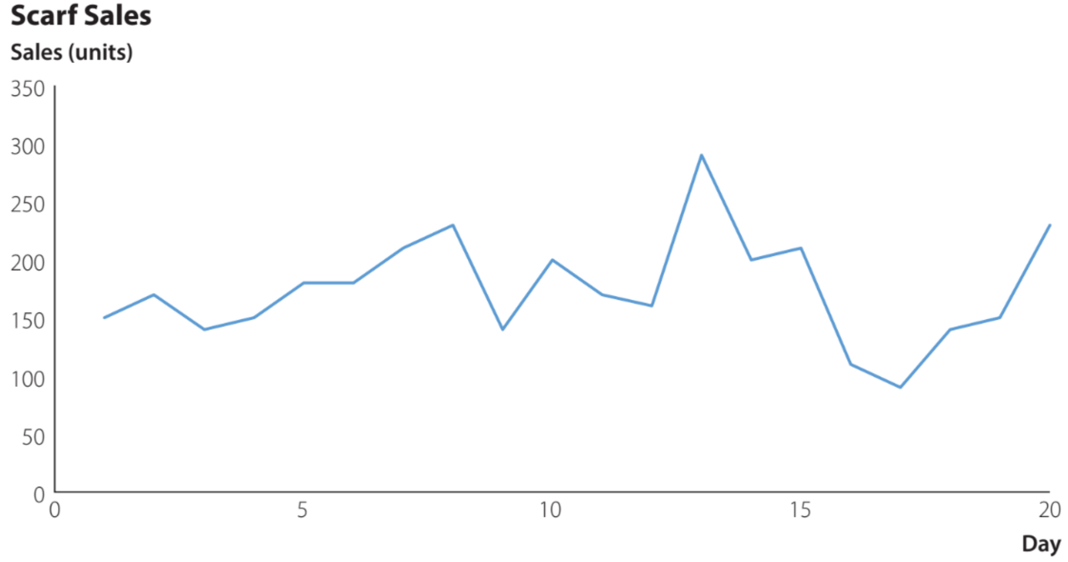
	Month						
	1	2	3	4	5	6	Total
Costs ($)	48,123	56,458	64,125	52,158	54,718	50,985	326,567
Revenues ($)	64,124	66,128	67,125	48,178	51,785	55,687	353,027
Profits ($)	16,001	9,670	3,000	(3,980)	(2,933)	4,702	26,460

Increase Data-Ink Ratio in Charts to Increase Clarity

Example of low data-ink chart

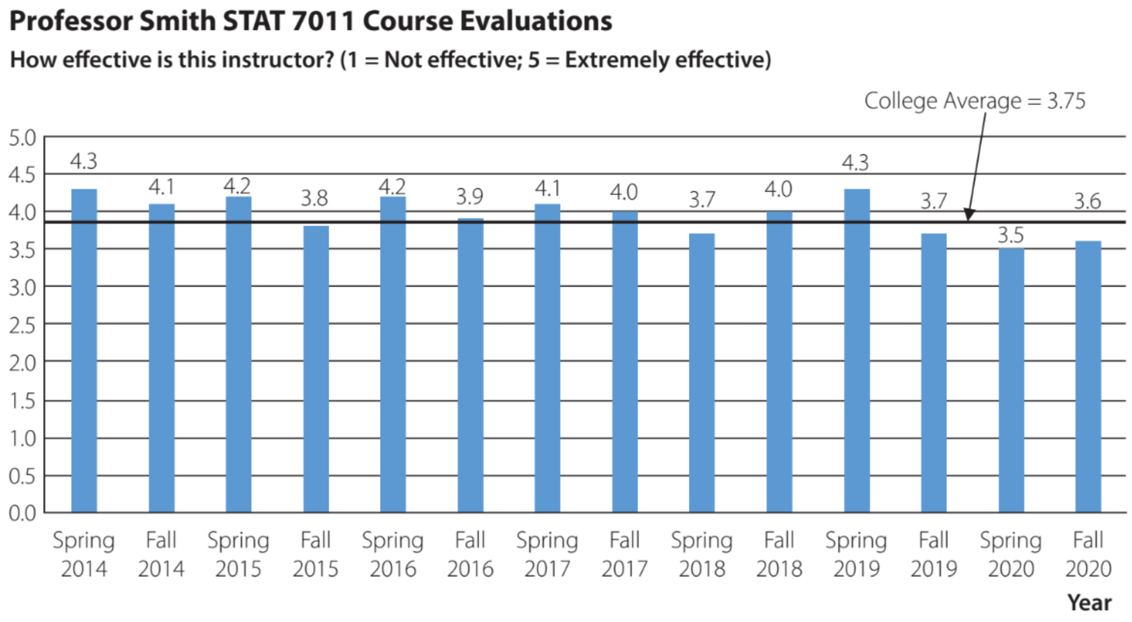


Example of high data-ink chart

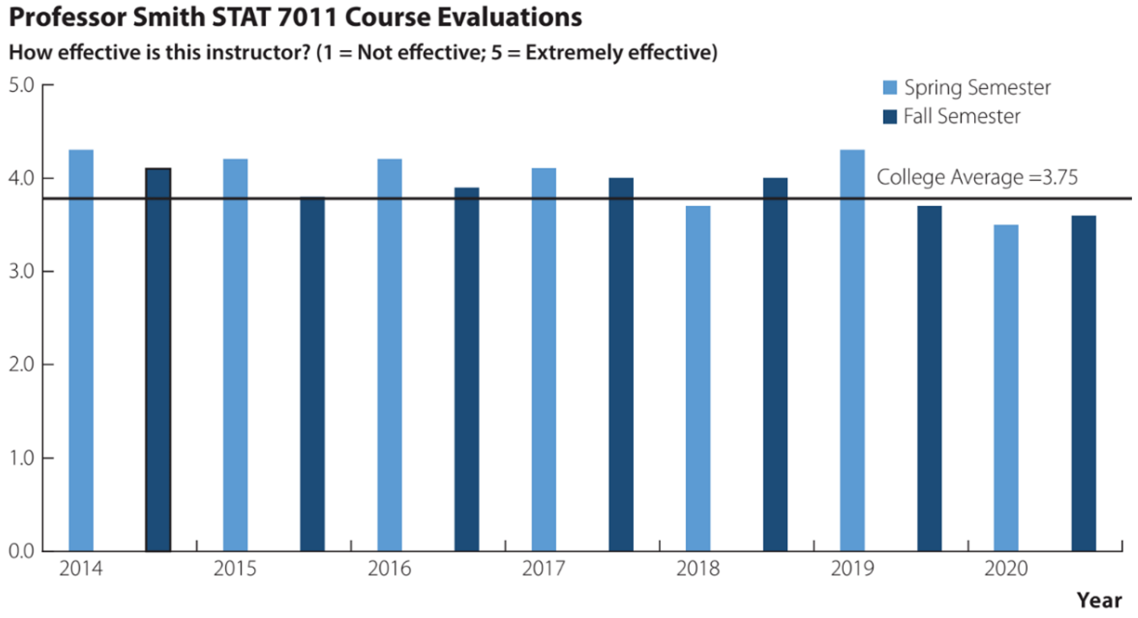


Increase Data-Ink Ratio in Charts to Decrease Clutter

A cluttered column chart for the STAT 7011 course evaluation data



An improved column chart for the STAT 7011 course evaluation data



Suggestions for Text Alignment in Tables

- Columns of numerical values in a table should usually be right-aligned.
- All columns showing values should include the same number of decimal digits.
- Only include the necessary number of decimal digits. Additional digits that are not meaningful to the audience increase clutter.
- For extremely large numbers, display data rounded to the nearest thousand, ten thousand, or million.
- It is generally most effective to left-align text values within a column in a table.

End of Part 2

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